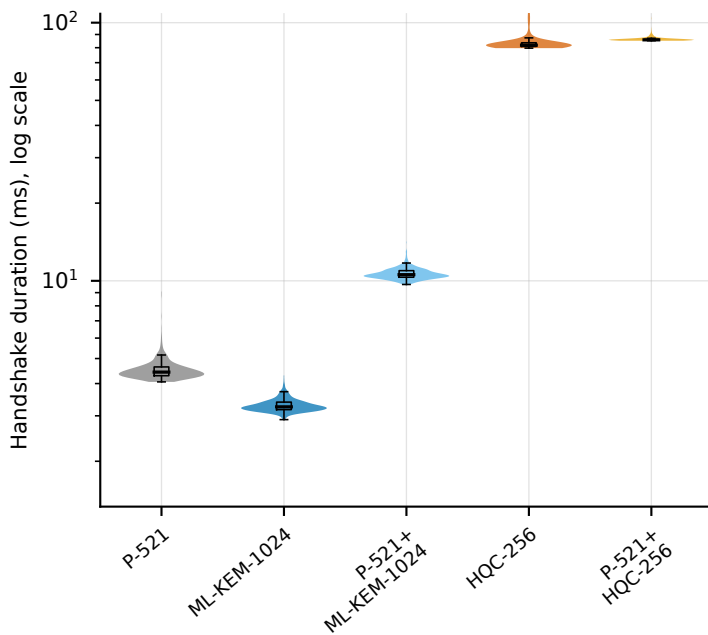
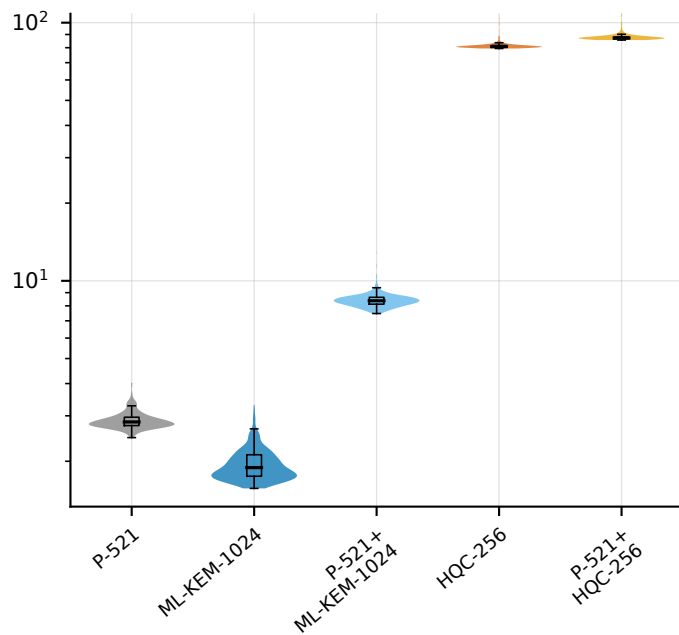


(a)

secp521r1 (classical)

**(b)**

mlds87 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mlds87 Mean (ms)	mlds87 CV (%)	mlds87 Max (ms)
P-521	4.55	10.1	9.09	2.89	7.9	4.03
ML-KEM-1024	3.30	5.8	4.30	1.97	14.8	3.31
P-521+ML-KEM-1024	10.7	5.0	14.2	8.40	5.9	13.0
HQC-256	84.6	10.7	140.0	81.3	2.9	106.0
P-521+HQC-256	86.4	3.1	117.8	88.0	4.5	141.6